

MATLAB-BASED MODELLING OF VSB MODULATOR AND DEMODULATOR FOR ANALOG TELEVISION COMMUNICATION APPLICATIONS

AIM :

To generate and analyse the Vestigial Sideband (VSB) modulated signal by performing DSB-SC modulation and applying a suitable filter using MATLAB. ii) To implement coherent demodulation and recover the original message signal from the VSB signal, and compare the transmitted and recovered waveforms.

Program code:

% Experiment 3: VSB modulation and coherent demodulation

clear; clc; close all;

% Parameters

Fs = 100000; % Sampling frequency (Hz)

T = 0.05; % Duration (s)

t = 0:1/Fs:T-1/Fs;

fc = 10000; % Carrier frequency (Hz)

fm = 500; % Message frequency (Hz)

Ac = 1; % Carrier amplitude

Am = 0.5; % Message amplitude

message = Am*cos(2*pi*fm*t);

amSignal = (Ac + message).*cos(2*pi*fc*t);

% Design an asymmetric bandpass filter to approximate VSB:

% retain the upper sideband and a vestige of the lower sideband

```
bp = designfilt('bandpassfir', ...  
    'FilterOrder', 600, ...  
    'CutoffFrequency1', fc-300, ...  
    'CutoffFrequency2', fc+1000, ...  
    'SampleRate', Fs, ...  
    'Window', 'hamming');
```

```
vsbSignal = filter(bp, amSignal);
```

% Coherent demodulation

```
mixedSignal = 2 * vsbSignal .* cos(2*pi*fc*t);
```

% Low-pass filter to recover the message

```
lp = designfilt('lowpassfir', ...  
    'FilterOrder', 500, ...  
    'CutoffFrequency', 1500, ...  
    'SampleRate', Fs, ...  
    'Window', 'hamming');
```

```
recovered = filter(lp, mixedSignal);
```

% Remove approximate filter delay for clearer comparison

```
delay = round((600 + 500)/2);  
recoveredAligned = [recovered(delay+1:end), zeros(1,delay)];
```

```

% Plot signals

figure;
subplot(3,1,1);
plot(t, message);
title('Original Message');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(3,1,2);
plot(t, vsbSignal);
title('VSB Modulated Signal');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(3,1,3);
plot(t, recoveredAligned);
title('Recovered Message (Coherent Demodulation)');
xlabel('Time (s)'); ylabel('Amplitude'); grid on;

% Display spectra

N = length(t);
f = (-N/2:N/2-1)*(Fs/N);

figure;
subplot(2,1,1);
plot(f, abs(fftshift(fft(amSignal)))/N);
xlim([fc-2000 fc+2000]);
title('AM Spectrum Near Carrier');

```

```
xlabel('Frequency (Hz)'); ylabel('Magnitude'); grid on;
```

```
subplot(2,1,2);
```

```
plot(f, abs(fftshift(fft(vsbSignal)))/N);
```

```
xlim([fc-2000 fc+2000]);
```

```
title('VSB Spectrum Near Carrier');
```

```
xlabel('Frequency (Hz)'); ylabel('Magnitude'); grid on;
```

OUTPUT :



